



Using the %Dictionary Classes

Version 2026.1
2026-04-20

Using the %Dictionary Classes

PDF generated on 2026-04-20

InterSystems IRIS Version 2026.1

Copyright © 2026 InterSystems Corporation

All rights reserved.

InterSystems®, HealthShare Care Community®, HealthShare Unified Care Record®, InterSystems IRIS® IntegratedML®, InterSystems Caché®, InterSystems Ensemble®, InterSystems HealthShare®, InterSystems IRIS®, InterSystems IRIS® for Health, and TrakCare are registered trademarks of InterSystems Corporation. HealthShare® Health Connect Cloud™, InterSystems® Data Fabric Studio™, and InterSystems Supply Chain Orchestrator™ are trademarks of InterSystems Corporation. TrakCare is a registered trademark in Australia and the European Union.

All other brand or product names used herein are trademarks or registered trademarks of their respective companies or organizations.

This document contains trade secret and confidential information which is the property of InterSystems Corporation, One Congress Street, Boston, MA 02114, or its affiliates, and is furnished for the sole purpose of the operation and maintenance of the products of InterSystems Corporation. No part of this publication is to be used for any other purpose, and this publication is not to be reproduced, copied, disclosed, transmitted, stored in a retrieval system or translated into any human or computer language, in any form, by any means, in whole or in part, without the express prior written consent of InterSystems Corporation.

The copying, use and disposition of this document and the software programs described herein is prohibited except to the limited extent set forth in the standard software license agreement(s) of InterSystems Corporation covering such programs and related documentation. InterSystems Corporation makes no representations and warranties concerning such software programs other than those set forth in such standard software license agreement(s). In addition, the liability of InterSystems Corporation for any losses or damages relating to or arising out of the use of such software programs is limited in the manner set forth in such standard software license agreement(s).

THE FOREGOING IS A GENERAL SUMMARY OF THE RESTRICTIONS AND LIMITATIONS IMPOSED BY INTERSYSTEMS CORPORATION ON THE USE OF, AND LIABILITY ARISING FROM, ITS COMPUTER SOFTWARE. FOR COMPLETE INFORMATION REFERENCE SHOULD BE MADE TO THE STANDARD SOFTWARE LICENSE AGREEMENT(S) OF INTERSYSTEMS CORPORATION, COPIES OF WHICH WILL BE MADE AVAILABLE UPON REQUEST.

InterSystems Corporation disclaims responsibility for errors which may appear in this document, and it reserves the right, in its sole discretion and without notice, to make substitutions and modifications in the products and practices described in this document.

For Support questions about any InterSystems products, contact:

InterSystems Worldwide Response Center (WRC)

Tel: +1-617-621-0700

Tel: +44 (0) 844 854 2917

Email: support@InterSystems.com

Table of Contents

Using the %Dictionary Classes.....	1
1 Introduction to Class Definition Classes	1
2 Browsing Class Definitions	2
3 Modifying Class Definitions	3
4 See Also	3

Using the %Dictionary Classes

This topic discusses the *class definition classes*, a set of persistent classes that provide object and SQL access to all class definitions.

1 Introduction to Class Definition Classes

The class definition classes provide object and SQL access to all class definitions. Using these classes, you can programmatically examine class definitions, modify class definitions, create new classes, and even write programs that automatically generate documentation. These classes are contained within the %Dictionary package.

Note: There is an older set of class definition classes defined within the %Library package. These are maintained for compatibility with existing applications. New code should make use of the classes within the %Dictionary package. Make sure that you specify the correct package name when using these classes or you may inadvertently use the wrong class.

There are two parallel sets of class definition classes: those that represent *defined* classes and those that represent *compiled* classes.

A *defined* class definition represents the definition of a specific class. It includes only information defined by that class; it does not include information inherited from superclasses. In addition to providing information about classes in the dictionary, these classes can be used to programmatically alter or create new class definitions.

A *compiled* class definition includes all of the class members that are inherited from superclasses. A compiled class definition object can only be instantiated from a class that has been compiled. You cannot save a compiled class definition.

This page discusses defined class definitions exclusively, though the operation of the compiled class definitions is similar.

The family of class definition classes that represent *defined classes* includes:

Class	Description
%Dictionary.ClassDefinition	Represents a class definition. Contains class keywords as well as collections containing class member definitions.
%Dictionary.ForeignKeyDefinition	Represents a foreign key definition within a class.
%Dictionary.IndexDefinition	Represents an index definition within a class.
%Dictionary.MethodDefinition	Represents a method definition within a class.
%Dictionary.ParameterDefinition	Represents a parameter definition within a class.
%Dictionary.PropertyDefinition	Represents a property definition within a class.
%Dictionary.QueryDefinition	Represents a query definition within a class.
%Dictionary.TriggerDefinition	Represents an SQL trigger definition within a class.

Important: To reiterate, the content of an uncompiled class definition (as an instance of the `%Dictionary.ClassDefinition`) is not necessarily the same as the content of a compiled class definition (as an instance of `%Dictionary.CompiledClass`). The `%Dictionary.ClassDefinition` class provides an API to inspect or change the definition of the class — it does not ever represent the compiled class with inheritance resolved; `%Dictionary.CompiledClass`, on the other hand, *does* represent the compiled class with inheritance resolved.

For example, if you are trying to determine the value of a particular keyword in a class definition, use the `keywordnameIsDefined()` method from `%Dictionary.ClassDefinition` (such as **`OdbcTypeIsDefined()`** or **`ServerOnlyIsDefined()`**). If this boolean method returns false, then the keyword is not explicitly defined for the class. If you check the value of the keyword for the class definition, it will be the default value. However, after compilation (which includes inheritance resolution), the value of the keyword is determined by inheritance and may differ from the value as defined.

2 Browsing Class Definitions

You can use the SQL pages of the Management Portal to browse the class definition classes.

Similarly, you can programmatically browse through the class definitions using the same techniques you would use to browse any other kind of data: you can use dynamic SQL and you can instantiate persistent objects that represent specific class definitions.

For example, from within an InterSystems IRIS® data platform process, you can get a list of all classes defined within the dictionary for the current namespace by using the **`%Dictionary.ClassDefinition:Summary()`** query:

ObjectScript

```
set stmt=##class(%SQL.Statement).%New()
set status = stmt.%PrepareClassQuery("%Dictionary.ClassDefinition","Summary")
if $$$ISERR(status) {write "%Prepare failed:" do $SYSTEM.Status.DisplayError(status) quit}

set rset=stmt.%Execute()
if (rset.%SQLCODE != 0) {write "%Execute failed:", !, "SQLCODE ", rset.%SQLCODE, ": ", rset.%Message
quit}

while rset.%Next()
{
    write rset.%Get("Name"), !
}
if (rset.%SQLCODE < 0) {write "%Next failed:", !, "SQLCODE ", rset.%SQLCODE, ": ", rset.%Message
quit}
```

This sample method will write the names of all the classes visible in the current namespace (including classes in the system library). You can filter out unwanted classes using the various columns returned by the **`%Dictionary.ClassDefinition:Summary()`** query.

You can get detailed information about a specific class definition by opening a `%Dictionary.ClassDefinition` object for the class and observing its properties. The ID used to store `%Dictionary.ClassDefinition` objects is the class name:

ObjectScript

```
Set cdef = ##class(%Dictionary.ClassDefinition).%OpenId("Sample.Person")
Write cdef.Name, !

// get list of properties
Set count = cdef.Properties.Count()
For i = 1:1:count {
    Write cdef.Properties.GetAt(i).Name, !
}
```

Note that you must fully qualify class names with their package name or the call to **`%OpenId()`** will fail.

3 Modifying Class Definitions

You can modify an existing class definition by opening a `%Dictionary.ClassDefinition` object, making the desired changes, and saving it using the `%Save()` method.

You can create a new class by creating a new `%Dictionary.ClassDefinition` object, filling in its properties and saving it. When you create `%Dictionary.ClassDefinition` object, you must pass the name of the class via the `%New()` command. When you want to add a member to the class (such as a property or method), you must create the corresponding definition class (passing its `%New()` command a string containing "class_name.member_name") and add the object to the appropriate collection within the `%Dictionary.ClassDefinition` object.

For example:

ObjectScript

```
Set cdef = ##class(%Dictionary.ClassDefinition).%New("MyApp.MyClass")
If $SYSTEM.Status.IsError(cdef) {
    Do $system.Status.DecomposeStatus(%objlasterror, .Err)
    Write !, Err(Err)
}
Set cdef.Super = "%Persistent,%Populate"

// add a Name property
Set pdef = ##class(%Dictionary.PropertyDefinition).%New("MyClass:Name")
If $SYSTEM.Status.IsError(pdef) {
    Do $system.Status.DecomposeStatus(%objlasterror, .Err)
    Write !, Err(Err)
}

Do cdef.Properties.Insert(pdef)

Set pdef.Type="%String"

// save the class definition object
Do cdef.%Save()
```

4 See Also

- [Defining Classes](#)
- [Top Level Class Syntax and Keywords](#)

